

VINAYAK GARG

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PROFESSIONAL SUMMARY

AI/ML Engineer with hands-on experience in end-to-end machine learning pipelines, deep learning, and NLP. Proficient in Python, PyTorch, TensorFlow, and Scikit-learn. Skilled in feature engineering, statistical analysis, model evaluation, and productionising models via Docker, FastAPI, and CI/CD pipelines on AWS. Strong foundation in data pipelines, experiment tracking (MLflow), and generative AI / LLM-based systems.

TECHNICAL SKILLS

Languages: Python, SQL, C++, Java, HTML/CSS

ML / Data Science: Scikit-learn, Pandas, NumPy, Feature Engineering, Statistical Analysis, Hyperparameter Tuning, Model Evaluation, A/B Testing

Deep Learning & NLP: PyTorch, TensorFlow, Transformers, GPT/LLM, Generative AI, NLTK, SpaCy, Hugging Face, TF-IDF, Word2Vec, BERT

MLOps & Infrastructure: Docker, FastAPI, Flask, MLflow, DagsHub, BentoML, AWS EC2/S3, GitHub Actions, CI/CD, Data Pipelines, REST APIs

Tools: Jupyter Notebook, VS Code, Google Colab, Streamlit, Postman, Git, PySpark, Matplotlib

PROJECTS

Website Phishing Detection System

Python | Docker | FastAPI | MLflow | DagsHub | AWS EC2/S3 | GitHub Actions

- Built production-grade ML pipeline using Python and Scikit-learn for real-time website phishing detection; achieved **F1=0.991**, Precision=0.990, Recall=0.992 with experiment tracking via MLflow on DagsHub.
- Designed multi-stage ETL data pipeline (ingestion → schema validation → transformation); containerised with Docker, deployed on AWS EC2/S3, and exposed via FastAPI REST API for scalable model inference.
- Enforced modular architecture with CI/CD via GitHub Actions enabling automated end-to-end retraining without manual intervention.

Customer Churn Prediction System

Python | Flask | Scikit-learn | Pandas | Feature Engineering | Statistical Analysis

- Built end-to-end churn prediction model using Python and Scikit-learn; achieved ~85% accuracy with 15% uplift through targeted feature engineering, hyperparameter tuning, and statistical model selection.
- Deployed Flask REST API enabling real-time predictions; model evaluation via precision, recall, F1-score, and confusion matrix.

Student Performance Prediction System

Python | Flask | XGBoost | CatBoost | GridSearchCV | Pandas | NumPy

- Benchmarked 8 regression algorithms (XGBoost, CatBoost, Random Forest) using GridSearchCV; best model achieved **R2=0.88**, RMSE=4.3, MAE=3.1 with train-test gap within 2%, confirming strong generalisation.
- Handled 12% missing values via median imputation, removed outliers beyond 3 SD; deployed via Flask UI for educators.

NLP: Sentiment Analysis & Spam Classification

Python | NLTK | SpaCy | TF-IDF | Word2Vec | Hugging Face Transformers

- Built NLP pipeline using Python, NLTK, SpaCy with tokenisation, stopword removal, TF-IDF and Word2Vec; developed spam classifier with AvgWord2Vec and Transformer-based features; evaluated via Precision, Recall, F1, Confusion Matrix.

EXPERIENCE

Technical Team Member — CSI-Innowave, MAIT Delhi

Sep 2024 – Present

- Co-organised Zephyr tech fest (100+ attendees); managed registrations, logistics, and post-event documentation.
- Co-facilitated technical ML/AI workshops and coordinated cross-functional sub-teams on data-driven projects.

EDUCATION

B.Tech – Computer Science & Engineering | MAIT Delhi

CGPA: 7.96/10 | Expected 2028

Class XII – CBSE | SGN Public School, Delhi

92% | 2024

Class X – CBSE | Seth Anandram Jaipuria School, Vasundhara

96% | 2022

ACHIEVEMENTS

- JEE Mains — 96 Percentile (top 4% nationally out of 1 million+ candidates).
- Solved 100+ DSA problems on competitive coding platforms; scored full marks in Artificial Intelligence.
- Self-driven AI/ML learner: independently built and deployed four end-to-end ML/NLP/MLOps projects beyond coursework.